

insights. Farmers can make data-driven decisions about planting, harvesting, and resource allocation.

Remote monitoring: Farmers can remotely monitor their farms using smartphones or computers. This enables them to respond quickly to issues and make adjustments to farm operations from anywhere.

Resource efficiency: IoT helps reduce resource wastage by optimising the use of water, fertilisers, pesticides, and energy. This not only reduces costs but also minimises the environmental impact of agriculture.

Market access: IoT can provide farmers with access to real-time market information, helping them make informed decisions about when and where to sell their products to get the best prices.

Sustainability: IoT-enabled precision agriculture practices can contribute to sustainable farming by reducing environmental impact, conserving resources, and minimising the use of chemicals.

Drones as IoT devices

Drones, also known as unmanned aerial vehicles (UAVs), are increasingly being used as IoT devices in agriculture to gather data, monitor crops, and make informed decisions for improved farming practices.

Equipped with cameras and sensors, drones can fly over agricultural fields to capture high-resolution images and data about crop health. These images can be analysed using IoT technology to detect early signs of disease, nutrient deficiencies, or pest infestations. This real-time monitoring allows farmers to take prompt action to mitigate issues.

Drones are integrated with GPS technology and IoT sensors to precisely measure and map fields. This information is used to optimise irrigation, fertilisation, and pesticide application, reducing resource wastage and improving crop yields.

Drones can carry sensors that measure soil moisture, pH levels, and nutrient content. This data helps to make data-driven decisions about soil management, such as when and where to irrigate or apply fertilisers.

Drones can provide real-time weather data, including temperature, humidity, wind speed, and precipitation levels. This information helps farmers make informed decisions about planting, harvesting, and protecting crops from adverse weather conditions. They can also be used to monitor and manage livestock. IoT-enabled drones can track the location of animals, assess their health, and even provide real-time video feeds for remote monitoring.

Drones can also be equipped with sprayers to apply pesticides or fertilisers precisely where needed. IoT technology controls and optimises the spraying process based on data from sensors, reducing chemical usage and minimising environmental impact.

All the data collected by drones can be sent to cloud-based IoT platforms for analysis. Farmers and agronomists can access this data remotely to make data-driven decisions, track trends over time, and create predictive models for better crop management.

Using historical data, drones can help predict crop yields for a season. This information is crucial for crop planning, resource allocation, and marketing strategies.

In case of natural disasters or crop damage, drones can be used to assess the extent of the damage quickly and accurately.

IoT and soil sensors

The use of soil probes as IoT devices in agriculture enhances precision farming, resource efficiency, and environmental sustainability. It empowers farmers to make informed decisions, optimise crop management, assess soil conditions, and boost yields while conserving water and fertilisers.

These IoT-equipped probes, armed with sensors for soil moisture, temperature, pH, and nutrient levels, continually gather real-time field data. This data is wirelessly transmitted to a central control system or cloud-based platform via technologies like Wi-Fi, cellular networks, or LoRaWAN. Farmers and agronomists access this data through web or mobile apps, enabling remote monitoring and timely decision-making.

Soil moisture sensors on these probes relay information about soil moisture levels at varying depths, allowing automated irrigation systems to trigger irrigation only when necessary, thus preventing water wastage.

IoT-enabled soil probes, equipped with nutrient sensors, assess soil nutrient levels and recommend precise fertiliser application rates, reducing excess fertiliser use and nutrient runoff. Integration of these sensors with weather forecasting services enables farmers to make informed decisions based on upcoming weather events, influencing planting and harvesting schedules.

Moreover, these IoT-based soil probes can be integrated with other farm equipment and actuators, granting farmers remote control over irrigation systems, valves, and devices based on real-time soil data, weather forecasts, and preset conditions.

The probes also send alerts and notifications to farmers' smartphones or email addresses when soil conditions deviate from desired levels, facilitating rapid responses to issues such as soil moisture depletion or pH imbalances.

Furthermore, IoT systems store historical soil data, allowing for trend analysis and identification of patterns. This historical data informs strategies for long-term soil health improvement, crop rotation planning, and farming practice optimisation.